

ORGAN SYSTEMS

Digestive System

The digestive system consists of **three divisions**:

mouth, pharynx, and the **alimentary canal** (esophagus, stomach, small intestine, large intestine).

- **Mouth**: food is moistened by saliva and mechanically processed by the **teeth, tongue, and mandible**.
 - **Pharynx and esophagus**: propel food into the stomach.
 - **Stomach**: mixes food with **gastric juice**, converting it into **chyme**.
 - **Small intestine**: enzymatic digestion by **intestinal glands, liver, gallbladder, and pancreas**; digested nutrients are absorbed into **blood and lymph capillaries**.
 - **Large intestine**: absorbs **water and electrolytes**; waste is compacted and eliminated via the **rectum and anal canal** as feces.
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Respiratory System

The respiratory system has a **conducting portion** and a **respiratory portion**.

- **Conducting portion**: nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, and bronchi.
 - Air is **filtered, humidified, and warmed** by mucous membranes.
 - **Respiratory portion**: lungs containing **alveoli**, where **oxygen–carbon dioxide exchange** occurs with assistance from the **diaphragm and thoracic cage**.
 - **Speech**: produced by intermittent airflow and **opening and closing of the glottis**.
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Urinary System

The urinary system includes the **kidneys, ureters, urinary bladder, and urethra**.

- **Kidneys**: filter blood, remove wastes, and produce urine.
- **Ureters**: transport urine to the bladder.
- **Bladder**: stores urine.
- **Urethra**: conveys urine out of the body.

Additional kidney functions:

- Regulation of **water and electrolyte balance**
 - Maintenance of **acid–base balance**
 - Control of **urine volume and composition**
 - Regulation of **blood volume and blood pressure**
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Reproductive System

A. Male Reproductive System

Components include:

- **Testes:** produce spermatozoa and sex hormones
- **Duct system:** epididymis → ductus deferens → ejaculatory duct → urethra
- **Accessory glands:** seminal vesicles, prostate, bulbourethral glands (contribute to seminal fluid)
- **Urethra and penis:** convey ejaculate to the exterior

All accessory gland secretions enter the **urethra**.

B. Female Reproductive System

Includes **ovaries, uterine tubes, uterus, vagina, and external genitalia**.

- **Ovaries:** produce oocytes and steroid hormones.
- **Ovulation:** oocyte released into the pelvic peritoneal cavity.
- **Fimbriae** of uterine tubes capture the oocyte.
- **Fertilization** usually occurs in the **ampulla** of the uterine tube.
- **Implantation** occurs in the uterine wall.
- **Uterus + vagina** form the **birth canal**.
- **Vagina:** passage for delivery and receives penis and semen.

External genitalia:

- Mons pubis (fatty eminence)
 - Labia majora (large skin folds)
 - Labia minora (hairless, fat-free folds)
 - Vestibule (vaginal entrance with hymen)
 - Clitoris (erectile tissue with crura, body, glans; covered by prepuce)
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Endocrine System

A system of **ductless glands** that secrete **hormones directly into the bloodstream**.

- Integrates and regulates organ systems
- Plays a major role in **growth, metabolism, and reproduction**
- Acts more slowly than the nervous system

Pure endocrine glands:

- Pituitary, pineal, thyroid, parathyroid, adrenal

Organs with endocrine function:

- Pancreas, thymus, gonads, hypothalamus, kidneys, liver, stomach

Includes **tropic hormones**, which regulate other endocrine glands and physiological responses.

Integumentary System

Consists of the **skin** and its appendages: **sweat glands, sebaceous glands, hair, and nails**.

- Contains **sensory receptors** for pain, temperature, touch, and pressure.

A. Skin

- Largest organ of the body
- Layers:
 - **Epidermis**: avascular, keratinized stratified squamous epithelium (ectodermal origin), thickest on palms and soles
 - **Dermis**: connective tissue (mesodermal origin), contains hair follicles and glands
 - **Hypodermis**: deep fatty layer

Functions:

- Protection
- Sensation
- Temperature regulation
- Vitamin D production
- Absorption

B. Skin Appendages

- **Sweat glands**: excretion and thermoregulation
 - **Sebaceous glands**: secrete sebum into hair follicles, lubricating skin and hair
 - **Hair**: protection, temperature regulation, aids perspiration evaporation
 - **Nails**: epidermal thickenings protecting digit tips
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